

PRACTICAL NO.: 2

Experiment Title: Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

Software/ Tools Required: VS Code (text editor), HTML, CSS, JavaScript.

Theory

JavaScript is a high-level, lightweight, and interpreted programming language used to make web pages interactive and dynamic. It allows developers to create applications that can accept user input, perform calculations, validate data, and display results instantly without reloading the webpage. Modern JavaScript (ES6 and later) introduced several new features that make programming easier, more efficient, and easier to maintain.

JavaScript provides three keywords for declaring variables: var, let, and const.

var is the traditional way of declaring variables. It has function scope, which means it is accessible throughout the function in which it is declared. Variables declared with var can be redeclared and reassigned, which may sometimes cause unexpected behavior.

let was introduced in ES6 and has block scope. It can be reassigned but cannot be redeclared within the same block. It is commonly used for variables whose values change during program execution.

const is also block-scoped and is used for variables whose values should remain constant. A variable declared with const must be initialized at the time of declaration and cannot be reassigned or redeclared.

Modern JavaScript also introduces Template Literals, which are enclosed in backticks (```). Template literals allow variables and expressions to be embedded directly into strings using `${}`. They also support multi-line strings, making code more readable and easier to write.

Another important ES6 feature is Destructuring. Destructuring is used to extract values from arrays or properties from objects into separate variables in a simple and concise manner. It reduces the amount of code required and improves readability.

In this experiment, a Billing Calculator is created using HTML, CSS, and JavaScript. The user enters details such as product name, quantity, and price. JavaScript processes the input, calculates the total bill, and displays the final amount. The program demonstrates the use of var, let, and const for variable declaration, template literals for displaying formatted output, and destructuring for handling object or array data efficiently. This experiment helps in understanding modern JavaScript features and their practical implementation in web application development.

Result

The experiment was successfully performed. The billing calculator accepted user input, calculated the total bill correctly, and displayed the result on the webpage. The concepts of var, let, const, template literals, and destructuring were implemented successfully.

Output:

The screenshot displays the ShopNest online shopping interface. At the top, the ShopNest logo and tagline "Online Shopping Made Easy" are visible. Below the header, three product cards are shown: a Smart Phone (Latest Android Phone) priced at ₹20,000, Running Shoes (Comfortable Sports Shoes) priced at ₹3,000, and a Smart Watch (Fitness Smart Watch) priced at ₹5,000. Each card includes a quantity input field. The Smart Phone quantity is set to 3, Running Shoes to 2, and Smart Watch to 3. Below the product cards, a "Customer Details" form is shown with the Customer Name field filled with "GITIKA". A "Generate Bill" button is located below the form. The bottom section of the image shows the generated "Shopping Invoice". The invoice lists the items and their quantities: Smart Phone (3 x ₹20,000 = ₹60,000), Running Shoes (2 x ₹3,000 = ₹6,000), and Smart Watch (3 x ₹5,000 = ₹15,000). The subtotal is ₹81,000.00, and the GST (18%) is ₹14,580.00. The total bill is ₹95,580.00.

ShopNest
Online Shopping Made Easy

Smart Phone
₹20,000
Latest Android Phone
Quantity: 3

Running Shoes
₹3,000
Comfortable Sports Shoes
Quantity: 2

Smart Watch
₹5,000
Fitness Smart Watch
Quantity: 3

Customer Details
Customer Name: GITIKA

Generate Bill

Shopping Invoice

Customer Name: GITIKA
Phone Number: 253637 Gitika Makheja PRN: 24070521024

Smart Phone : 3 x ₹20000 = ₹60000
Running Shoes : 2 x ₹3000 = ₹6000
Smart Watch : 3 x ₹5000 = ₹15000

Subtotal: ₹81000.00
GST (18%): ₹14580.00
Total Bill: ₹95580.00

Conclusion

The experiment was completed successfully. It demonstrated the use of var, let, and const for variable declaration and explained the advantages of modern JavaScript features such as template literals and destructuring. The billing calculator helped in understanding user input handling, performing calculations, and displaying formatted output, thereby improving knowledge of modern JavaScript programming and web development.